

Integrating with exponentials and logs — 0.5.6/1.6

If we know that $\frac{d}{dx} F(x) = f(x)$

then we also know that $\int f(x) dx = F(x) + C$

Examples: $\frac{d}{dx} e^x = e^x \Rightarrow \boxed{\int e^x dx = e^x + C}$

$$\frac{d}{dx} a^x = a^x \ln(a) \Rightarrow \int a^x \ln(a) dx = a^x + C$$

$$\Rightarrow \boxed{\int a^x dx = \frac{a^x}{\ln(a)} + C}$$

$$\frac{d}{dx} \ln|x| = \frac{1}{x}$$

$$\Rightarrow \boxed{\int \frac{1}{x} dx = \ln|x| + C}$$

Example: Find $\int \frac{3x}{x^2+1} dx$.

$$u = x^2 + 1$$

$$du = 2x dx$$

$$dx = \frac{du}{2x}$$

$$= \int \frac{1}{u} \cdot 3x \cdot \frac{du}{2x} = \int \frac{1}{u} \cdot \frac{3}{2} du$$

$$= \frac{3}{2} \int \frac{1}{u} du = \frac{3}{2} \ln|u| + C$$

$$= \frac{3}{2} \ln|x^2+1| + C$$

Example: Find $\int \frac{\ln(x)}{x} dx$

$$u = \ln(x)$$

$$du = \frac{1}{x} dx$$

$$= \int u du = \frac{u^2}{2} + C = \frac{(\ln(x))^2}{2} + C$$

Example: $\int \tan(x) dx = \int \frac{\sin(x)}{\cos(x)} dx$

$$u = \cos(x)$$

$$du = -\sin(x) dx$$

$$= - \int \frac{1}{u} du = -\ln|u| + C$$

$$= -\ln|\cos(x)| + C$$

Example: $\int_0^2 \frac{e^x}{1+e^x} dx$

$$u = 1 + e^x$$

$$du = e^x dx$$

$$= \int_{u(0)}^{u(2)} \frac{1}{u} du = \int_2^{1+e^2} \frac{1}{u} du$$

$$= \ln|u| \Big|_2^{1+e^2} = \ln|1+e^2| - \ln|2| = \ln\left(\frac{1+e^2}{2}\right).$$